

HTML Input Types and Attributes

The `<input>` element creates a form control. Its **type** determines what the user can enter or select, while its **attributes** control details such as whether the field is required and what values are allowed.

```
<label for="email">Email:</label>
<input type="email" id="email" name="email" required>
```

Here, `type="email"` asks for an email address, `id` connects the field to its label, `name` identifies the submitted value, and `required` prevents an empty submission.

1. Common input types

Input type	Purpose	Example
text	Single-line text	<code><input type="text" name="student"></code>
password	Text hidden on screen	<code><input type="password" name="password"></code>
email	Email address	<code><input type="email" name="email"></code>
tel	Telephone number	<code><input type="tel" name="phone"></code>
url	Web address	<code><input type="url" name="website"></code>
search	Search query	<code><input type="search" name="query"></code>
number	Number with optional limits	<code><input type="number" name="age" min="18"></code>
range	Number selected using a slider	<code><input type="range" name="rating" min="1" max="10"></code>
date	Calendar date	<code><input type="date" name="visitDate"></code>
time	Time of day	<code><input type="time" name="visitTime"></code>
datetime-local	Date and time, without a time zone	<code><input type="datetime-local" name="meeting"></code>
month	Month and year	<code><input type="month" name="startMonth"></code>
week	Week and year	<code><input type="week" name="projectWeek"></code>
color	Colour picker	<code><input type="color" name="theme"></code>
file	File selection	<code><input type="file" name="assignment"></code>
radio	One choice from a group	<code><input type="radio" name="mode" value="online"></code>
checkbox	Independent yes/no choice or multiple choices	<code><input type="checkbox" name="agree" value="yes"></code>
hidden	Value without a visible control	<code><input type="hidden" name="formId" value="F01"></code>
submit	Submits the form	<code><input type="submit" value="Register"></code>
reset	Restores initial form values	<code><input type="reset" value="Clear"></code>
button	Button for a scripted action	<code><input type="button" value="Check"></code>

Input type	Purpose	Example
image	Image used as a submit button	<code><input type="image" src="send.png" alt="Submit"></code>

Note: `type="tel"` does not automatically validate a phone-number format. Use `pattern` if a specific format is required.

2. Core attributes

Attribute	Purpose	Example
type	Selects the kind of control	<code>type="email"</code>
name	Identifies the value in submitted form data	<code>name="studentEmail"</code>
id	Uniquely identifies the element; connects it to a label	<code>id="studentEmail"</code>
value	Sets an initial or submitted value, depending on input type	<code>value="yes"</code>
placeholder	Displays a temporary hint	<code>placeholder="name@example.com"</code>
required	Requires a value before submission	<code>required</code>
readonly	Prevents editing; value is normally submitted	<code>readonly</code>
disabled	Prevents use; value is not submitted	<code>disabled</code>
autocomplete	Gives the browser an autofill hint	<code>autocomplete="email"</code>
autofocus	Focuses the field when the page loads	<code>autofocus</code>

name versus id

```
<label for="rollNo">Roll number:</label>
<input type="text" id="rollNo" name="roll_number">
```

- `id="rollNo"` connects the label to the input.
- `name="roll_number"` is the key sent with the entered value.

A field without `name` is generally **not included** in submitted form data.

3. Validation attributes

Attribute	Applies mainly to	Purpose
minlength	Text-based inputs	Minimum number of characters
maxlength	Text-based inputs	Maximum number of characters
min	Number, range, date, time and related types	Lowest permitted value
max	Number, range, date, time and related types	Highest permitted value
step	Number, range, time and related types	Allowed increments
pattern	Text, tel, email, URL and related text inputs	Required text format
multiple	Email and file inputs	Allows multiple values or files

Attribute Applies mainly to

accept File input

Purpose

Suggests permitted file types

Examples

```
<!-- At least 8 characters -->
<input type="password" name="password" minlength="8" required>

<!-- Age from 18 to 25 -->
<input type="number" name="age" min="18" max="25" required>

<!-- 10 digits -->
<input type="tel" name="phone"
      pattern="[0-9]{10}"
      title="Enter exactly 10 digits">

<!-- Time between 9:00 AM and 5:00 PM -->
<input type="time" name="appointment"
      min="09:00" max="17:00">

<!-- PDF selection -->
<input type="file" name="assignment"
      accept=".pdf,application/pdf">
```

accept guides the file picker; the server must still verify the uploaded file. Likewise, browser validation helps users correct mistakes but must be followed by server-side validation.

4. Selecting options

Radio buttons: one selection

Radio buttons with the same name form one group.

```
<fieldset>
  <legend>Attendance mode</legend>

  <label>
    <input type="radio" name="mode" value="online" required>
    Online
  </label>

  <label>
    <input type="radio" name="mode" value="campus">
    On campus
  </label>
</fieldset>
```

Checkboxes: multiple selections

```
<fieldset>
  <legend>Topics of interest</legend>

  <label><input type="checkbox" name="topic" value="html"> HTML</label>
  <label><input type="checkbox" name="topic" value="css"> CSS</label>
  <label><input type="checkbox" name="topic" value="js"> JavaScript</label>
```

```
</fieldset>
```

A checked checkbox sends its `name` and `value`. An unchecked checkbox is normally omitted from the submitted data. For several checkboxes with the same `name`, the server must handle multiple values.

5. Slider with a displayed value

A range input creates a slider. Use `<output>` to show the value as it changes.

```
<label for="rating">Workshop rating:</label>
<input type="range" id="rating" name="rating"
      min="1" max="10" step="1" value="5"
      oninput="ratingValue.value = this.value">
<output id="ratingValue" for="rating">5</output>
```

The slider starts at 5 and lets the user select a whole number from 1 to 10.

6. Suggested values with `<datalist>`

`<datalist>` works with an input to offer suggestions. The user can still enter a different value.

```
<label for="city">City:</label>
<input type="text" id="city" name="city" list="cities">

<datalist id="cities">
  <option value="Chandigarh">
  <option value="Mohali">
  <option value="Panchkula">
</datalist>
```

The input's `list="cities"` matches the datalist's `id="cities"`. Use `<select>` instead if the user must choose only from a fixed set of options.

7. File input and form encoding

A file input needs an appropriate form setup for uploading:

```
<form action="upload.php" method="post" enctype="multipart/form-data">
  <label for="project">Project PDF:</label>
  <input type="file" id="project" name="project"
        accept=".pdf,application/pdf" required>
  <button type="submit">Upload</button>
</form>
```

`method` and `enctype` are attributes of `<form>`, while `type`, `name`, and `accept` here are attributes of `<input>`.